US 20250279672A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0279672 A1
(43) Pub. Date: Sep. 4, 2025
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(54) SYSTEMS AND METHODS FOR WORKSITE GEOGRAPHIC OPTIMIZATION

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(21) Appl. No.: **18/592,764**

(22) Filed: **Mar. 1, 2024**

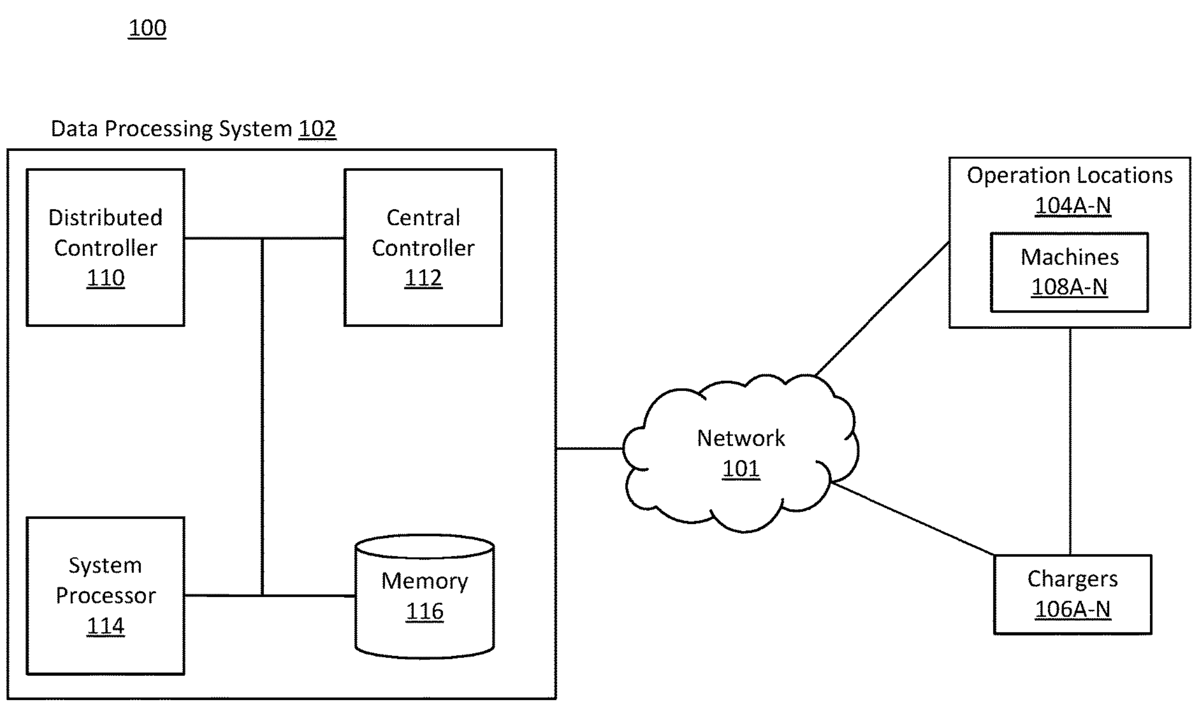
Publication Classification

(51) **Int. Cl.**
H02J 13/00 (2006.01)
H02J 7/00 (2006.01)

(52) **U.S. Cl.**
CPC ***H02J 13/00006*** (2020.01); ***H02J 7/00032*** (2020.01); ***H02J 7/0013*** (2013.01); ***H02J 7/0047*** (2013.01); ***H02J 7/00712*** (2020.01)

(57) **ABSTRACT**

Systems and methods for worksite geographic optimization may include one or more processors configured to calculate a worksite metric for a first worksite layout, the worksite metric based on a worksite cost and a historic worksite metric during a previous time period. In some embodiments, the first worksite layout differs from a second worksite layout. The processor(s) calculate a score corresponding to a machine at the first worksite layout, and select a charging location for the machine at the first worksite layout, according to the score and the worksite metric. The processor(s) can transmit a first signal corresponding to the charging location to the first machine, and a second signal to a charger of the plurality of chargers at the charging location.



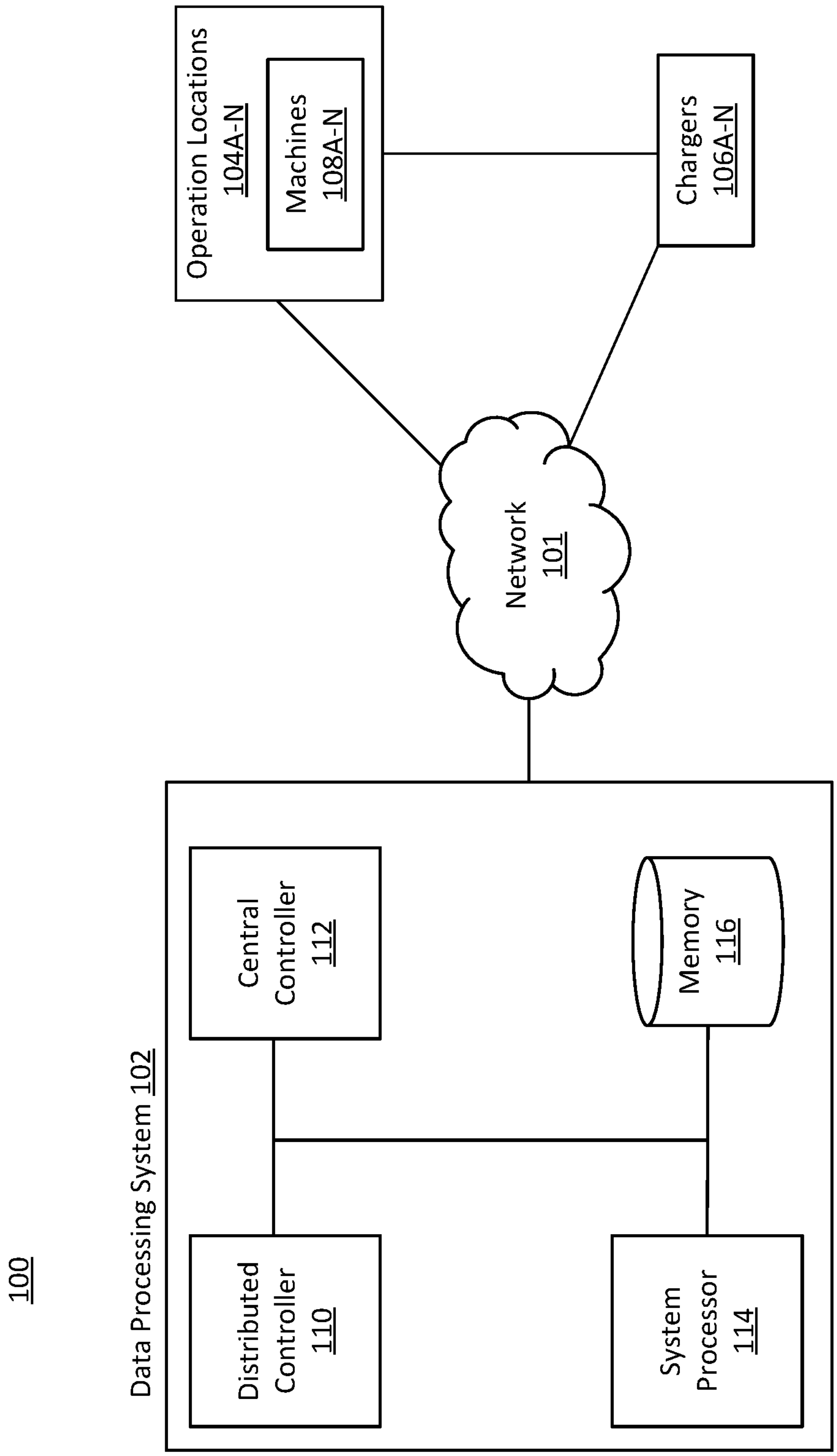

100
Data Processing System 102
Distributed Controller 110
Central Controller 112
System Processor 114
Memory 116
Network 101
Operation Locations 104A-N
Machines 108A-N
Chargers 106A-N

FIG. 1

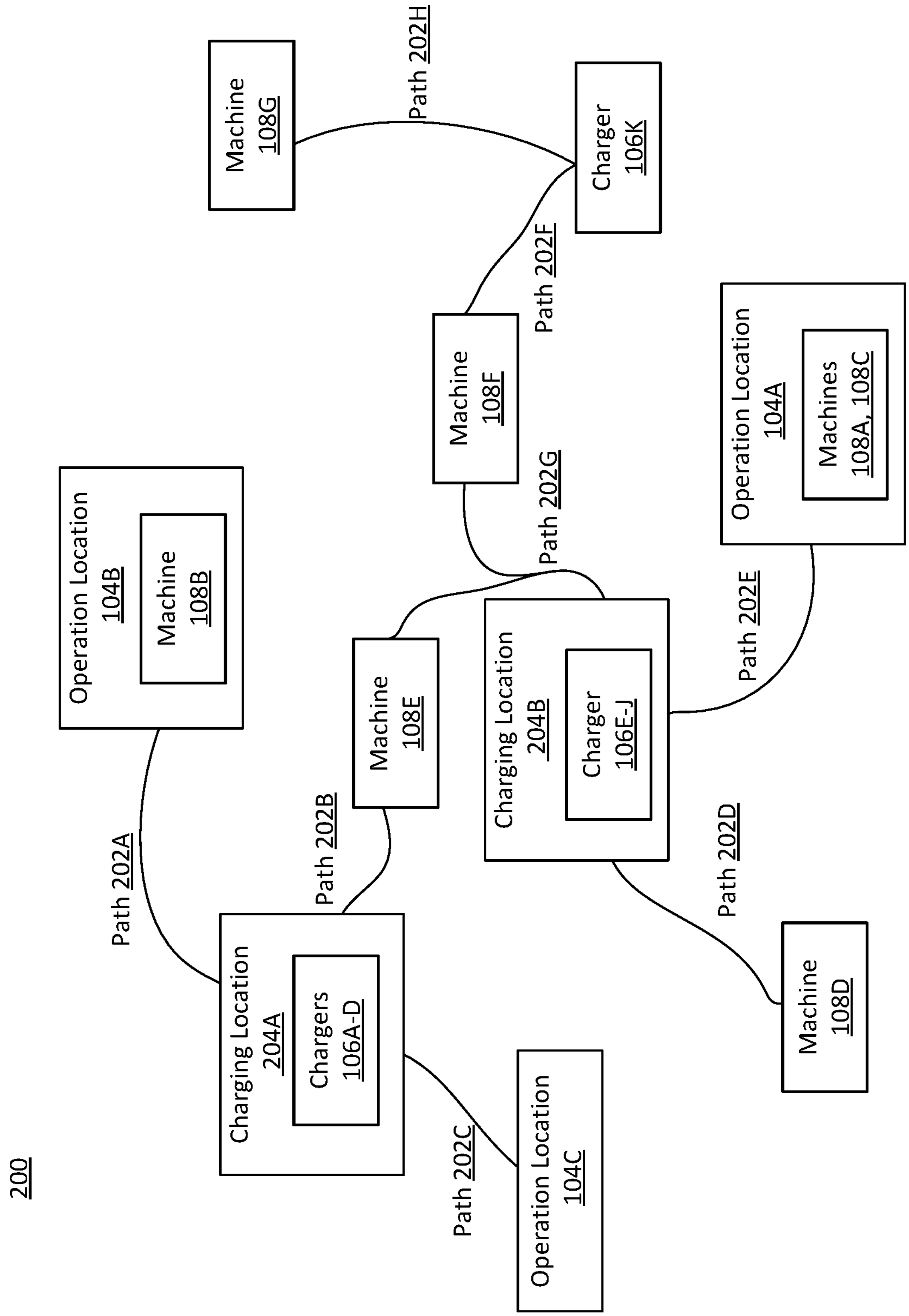

200
Machine 108G
Path 202H
Charger 106K
Path 202F
Machine 108F
Path 202G
Operation Location 104A
Machines 108A, 108C
Operation Location 104B
Machine 108B
Machine 108E
Charging Location 204B
Charger 106E-J
Path 202E
Path 202A
Path 202B
Path 202D
Charging Location 204A
Chargers 106A-D
Machine 108D
Path 202C
Operation Location 104C

FIG. 2

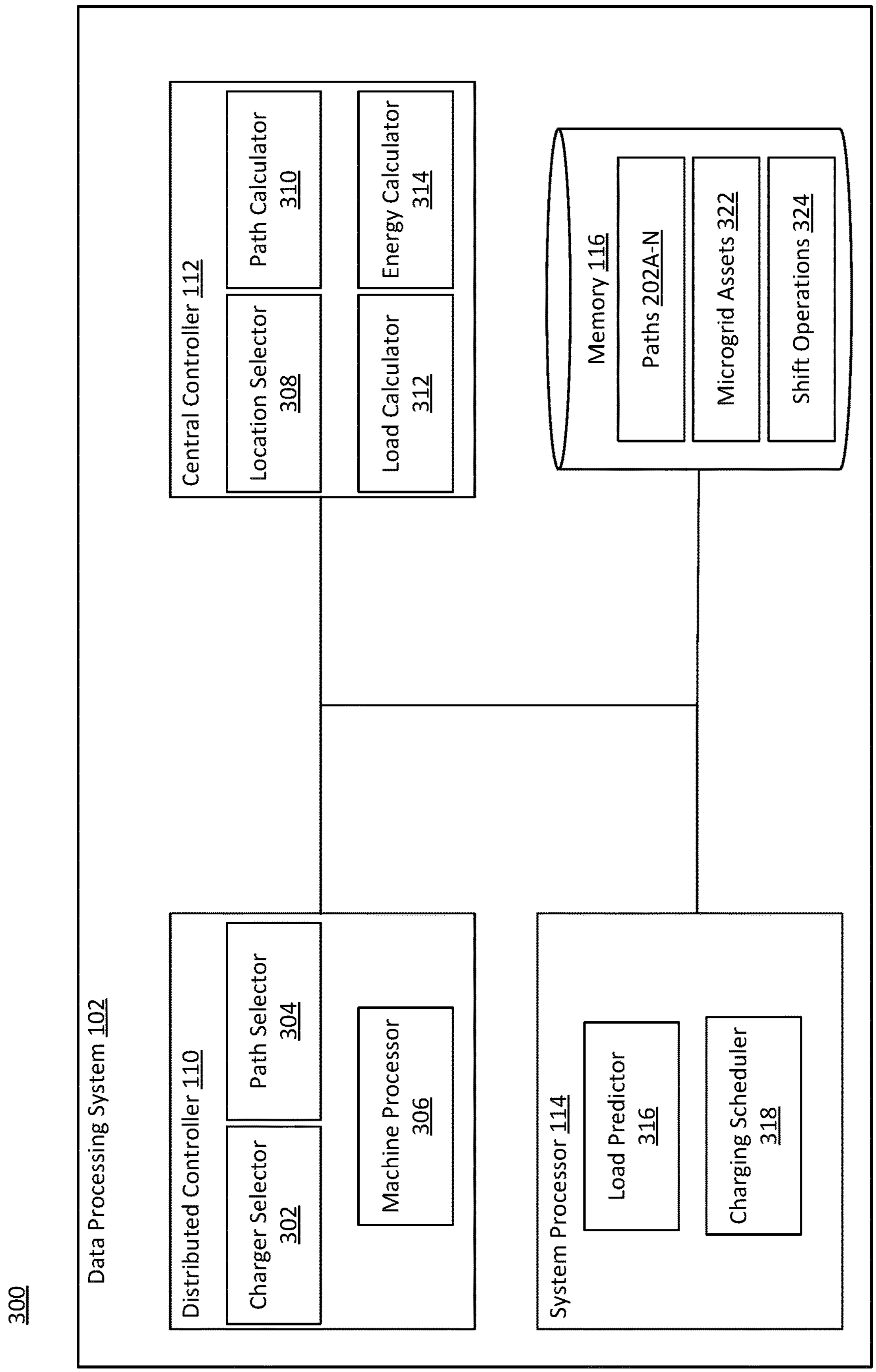

300
Data Processing System 102
Distributed Controller 110
Charger Selector 302
Path Selector 304
Machine Processor 306
System Processor 114
Load Predictor 316
Charging Scheduler 318
Central Controller 112
Location Selector 308
Path Calculator 310
Load Calculator 312
Energy Calculator 314
Memory 116
Paths 202A-N
Microgrid Assets 322
Shift Operations 324

FIG. 3

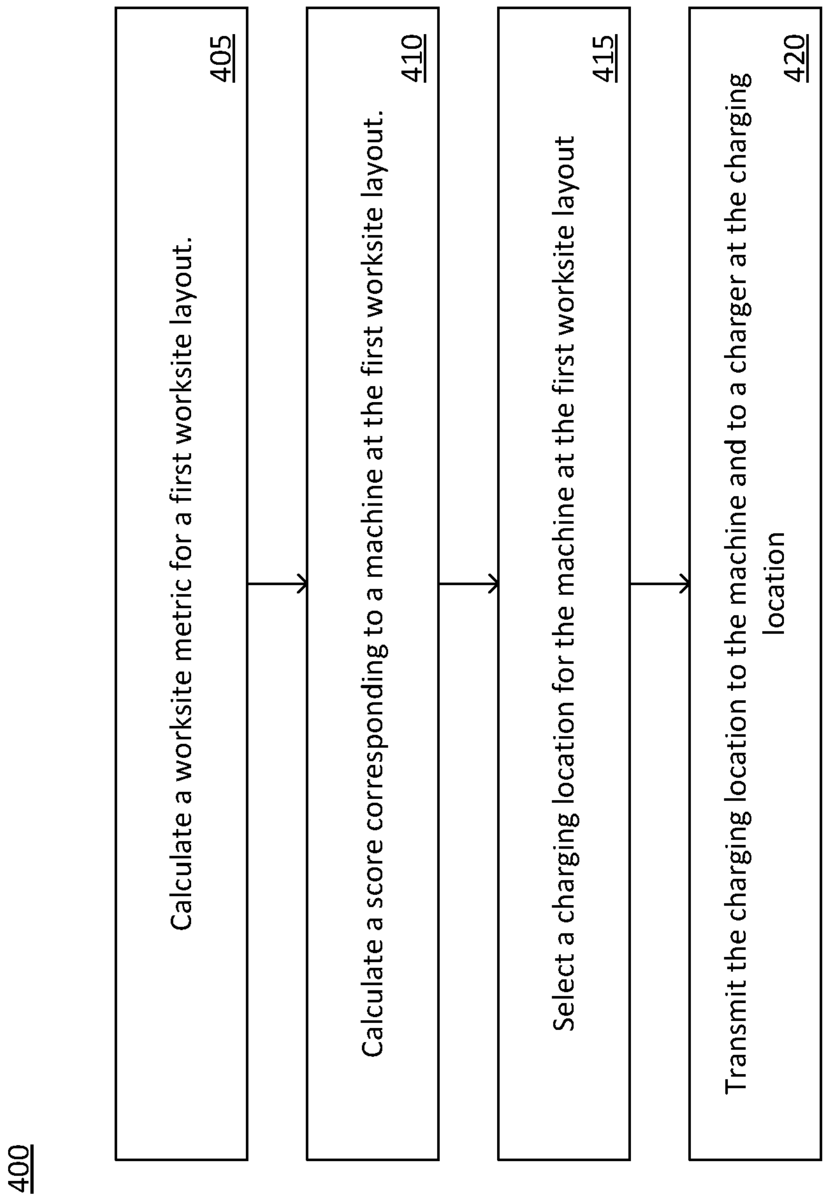

400
Calculate a worksite metric for a first worksite layout.
405
Calculate a score corresponding to a machine at the first worksite layout.
410
Select a charging location for the machine at the first worksite layout
415
Transmit the charging location to the machine and to a charger at the charging location
420

FIG. 4

SYSTEMS AND METHODS FOR WORKSITE GEOGRAPHIC OPTIMIZATION

TECHNICAL FIELD

[0001] The present implementations relate generally to optimizing locations of chargers at a worksite.

INTRODUCTION

[0002] The present disclosure relates generally to optimizing locations of chargers at a worksite. Worksites, such as construction sites, manufacturing plants, mining sites, and agricultural sites, may require machines to charge on the worksite.

SUMMARY

[0003] This disclosure is generally directed to optimizing locations of chargers at a worksite. With the introduction of electrified machines in a worksite, there is an increased need for optimally placed chargers at a worksite. By not moving chargers based on the worksite, issues of reduced productivity, increased energy demand, poor scheduling, increased project timelines and increase cost at the worksite arise. Aspects of this technical solution can provide a method for optimizing charger locations at a worksite to improve productivity, reduce energy demand, reduce scheduling issues, and reduce the cost at the worksite. For example, optimizing locations of chargers allows for a method that can include calculating, by one or more processors of a distributed controller, a worksite metric for a first worksite layout, the worksite metric based on a worksite cost and a historic worksite metric during a previous time period, wherein the first worksite layout differs from a second worksite layout. The method can include calculating, by the one or more processors of the distributed controller, a score corresponding to a machine at the first worksite layout. The method can include selecting, by the one or more processors of the distributed controller, a charging location for the machine at the first worksite layout, according to the score and the worksite metric. The method can include transmitting, by the one or more processors of the distributed controller, the charging location to the machine and to a charger at the charging location at the first worksite layout.

[0004] A first aspect provided herein relate to a method of worksite geographical optimization at a worksite. The method can include the one or more controllers calculating a worksite metric for a first worksite layout, the worksite metric based on a worksite cost and a historic worksite metric during a previous time period. In some embodiments, the first worksite layout differs from a second worksite layout. The method can include the one or more controllers calculating a score corresponding to a machine at the first worksite layout. The method can include the one or more controllers selecting a charging location for the machine at the first worksite layout, according to the score and the worksite metric. The method can include the one or more controllers transmitting a first signal corresponding to the charging location to the machine, and a second signal to a charger at the charging location.

[0005] In some embodiments, the method can include the one or more controllers calculating the worksite cost, based on a difference between a total cost and an amount of energy spent by the machine to reach the charger at the second worksite layout, during the previous time period. In some embodiments, the method can include the one or more controllers calculating the historic worksite metric based on a difference between a total amount of work and an amount of productive work at the second worksite layout, during the previous time period.

[0006] In some embodiments, the battery metrics can indicate at least one of a state of charge, a state of health, a voltage, charging time, or safety parameters. In some embodiments, the method can include the one or more controllers determining, the charging location based on the first worksite layout, movement of materials, and movement of the machine. In some embodiments, the method can include the one or more controllers determining the charger according to the charging location, and a path to the charger according to first worksite layout. In some embodiments, the method can include the one or more controllers transmitting a signal to the machine. In some embodiments, the signal can indicate the machine follow the path to the charger. In some embodiments, the signal can include information corresponding to load cycles and energy generation of the charge based on the previous time period.

[0007] In some embodiments, the method can include the one or more controllers receiving information between one or more machines and one or more chargers from a central controller. In some embodiments, the information can indicate battery metrics and charger availability. In some embodiments, the method can include the one or more controllers computing, using a linear optimizer, a new charger location for a future time period, according to the worksite cost and the historic worksite metric.

[0008] In some embodiments, the method can include the one or more controllers receiving a machine criticality at the first worksite layout corresponding to the machine, a queue at the charger, a charger connection success rate, and a time to charge based on a cost of energy from a central controller. In some embodiments, the method can include the one or more controllers calculating the score corresponding to the machine, according to the machine criticality, the queue at the charger, the charge connection success rate, and the time to charge.

[0009] In some embodiments, the method can include the one or more controllers sending the score to the charger and receiving an acceptance of the score by the charger. In some embodiments, the method can include the one or more controllers transmitting a signal to the machine. In some embodiments, the signal directs the machine to proceed to the charger of the received acceptance. In some embodiments, the method can include the one or more controllers broadcasting charger information of the charger of the received acceptance to a plurality of machines. In some embodiments, charger information can indicate an availability status and a power availability, each bid in the plurality of bids indicate battery metrics of a corresponding machine. In some embodiments, the method can include the one or more controllers receiving a plurality of bids from the plurality of machines and determining one bid in the plurality of bids corresponding to the charger information to charge the corresponding machine.

[0010] A second aspect provided herein relate to a system for worksite geographical optimization at the worksite. A system can include a plurality of machines. The plurality of machines can include one or more first machines at a first operation location and one or more second machines at a second operation location. The system can include a plurality of chargers. The plurality of chargers can include one or more first chargers at a first charging location and one or more second chargers at a second charging location. The system can include a data processing system. The data processing system can include memory and one or more controllers. The controller(s) can include one or more processors that can calculate a worksite metric for a first worksite layout. The worksite metric can be based on a worksite cost and a historic worksite metric during a previous time period. In some embodiments, the first worksite layout differs from a second worksite layout. The one or more processors can calculate a score corresponding to a first machine of the plurality of machines at the first worksite layout. The one or more processors can select a charging location for the machine at the first worksite layout, according to the score and the worksite metric. The one or more processors can transmit a first signal corresponding to the charging location to the first machine, and a second signal to a charger of the plurality of chargers at the charging location.

[0011] In some embodiments, a central controller of the one or more controllers can calculate the worksite cost, based on a difference between a total cost and an amount of energy spent by the first machine to reach the charger at the second worksite layout, during the previous time period. In some embodiments, the central controller can calculate the historic worksite metric based on a difference between a total amount of work and an amount of productive work at the second worksite layout, during the previous time period.

[0012] In some embodiments, the battery metrics can indicate at least one of a state of charge, a state of health, a voltage, charging time, or safety parameters. In some embodiments, a distributed controller of the one or more controllers can determine the charging location based on the first worksite layout, movement of materials, and movement of the first machine. In some embodiments, the distributed controller can determine the charger according to the charging location and a path to the charger according to the first worksite layout. In some embodiments, the distributed controller can transmit a signal to the first machine. In some embodiments, the signal can instruct the first machine follow the path to the charger. In some embodiments, the signal can indicate load cycles and energy generation of the charger based on the previous time period.

[0013] In some embodiments, a distributed controller of the one or more controllers can receive information between one or more of the plurality of machines and one or more of the plurality of chargers from a central controller. In some embodiments, the information can indicate battery metrics and charger availability. In some embodiments, the one or more controllers can compute, using a linear optimizer, a new charger location for a future time period, according to the worksite cost and the historic worksite metric. In some embodiments, the one or more controllers can determine a third worksite layout based the new charger location.

[0014] In some embodiments, a distributed controller of the one or more controllers can receive a machine criticality at the first worksite layout corresponding to the first machine, a queue at the charger, a charger connection success rate, and a time to charge, and calculate the score corresponding to the first machine, according to the machine criticality, the queue at the charger, the charge connection success rate, and the time to charge.

[0015] In some embodiments, a distributed controller of the one or more controllers can send the score to the charger and receive an acceptance of the score by the charger. In some embodiments, the distributed controller can transmit the first signal to the first machine. In some embodiments, the first signal directs the machine to proceed to the charger of the received acceptance. In some embodiments, the distributed controller broadcast charger information of the charger of the received acceptance to a plurality of machines. In some embodiments, charger information can indicate an availability status and a power availability. In some embodiments, the distributed controller can receive a plurality of bids from the plurality of machines. In some embodiments, each bid in the plurality of bids can indicate battery metrics of a corresponding machine. In some embodiments, the distributed controller can determine one bid in the plurality of bids corresponding to the charger information to charge the corresponding machine.

[0016] In some embodiments, the one or more controllers include a central controller and a distributed controller. The distributed controller may calculate the worksite metric for the first worksite layout, and calculate the score corresponding to the first machine of the plurality of machines at the first worksite layout. The distributed controller may select the charging location for the first machine at the first worksite layout, according to the score and the worksite metric. The distributed controller may transmit the first signal corresponding to the charging location to the machine, and the second signal to the charger at the charging location.

[0017] In a third aspect, this disclosure is directed to a data processing system including memory and one or more controllers including one or more processors configured to calculate a worksite metric for a first worksite layout, the worksite metric based on a worksite cost and a historic worksite metric during a previous time period. The first worksite layout may differ from a second worksite layout. The controller(s) may be configured to calculate a score corresponding to a first machine of a plurality of machines at the first worksite layout. The controller(s) may be configured to select a charging location for the first machine at the first worksite layout, according to the score and the worksite metric. The controller(s) may be configured to transmit a first signal corresponding to the charging location to the first machine, and a second signal to a charger of a plurality of chargers at the charging location.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] These and other aspects and features of the present implementations will become apparent to those ordinarily skilled in the art upon review of the following description of specific implementations in conjunction with the accompanying figures.

[0019] FIG. **1** depicts an example system to implement geography optimization at a worksite, in accordance with present implementations.

[0020] FIG. **2** depicts an example worksite layout, in accordance with present implementations.

[0021] FIG. **3** depicts an example data processing system to implement geography optimization at the worksite, in accordance with present implementations.

[0022] FIG. **4** is a flowchart showing an example method to implement geography optimization for the worksite, in accordance with present implementations.

DETAILED DESCRIPTION

[0023] This disclosure is generally directed to optimizing locations of chargers at a worksite. With the introduction of electrified machines in a worksite, there is an increased need for optimally placed chargers at a worksite. By not moving chargers based on the worksite, issues of reduced productivity, increased energy demand, poor scheduling, increased project timelines and increase cost at the worksite arise. Aspects of this technical solution can provide a method for optimizing charger locations at a worksite to improve productivity, reduce energy demand, reduce scheduling issues, and reduce the cost at the worksite. For example, optimizing locations of chargers allows for a method that can include calculating, by one or more processors, a worksite metric for a first worksite layout, the worksite metric based on a worksite cost and a historic worksite metric during a previous time period, wherein the first worksite layout differs from a second worksite layout. The method can include calculating, by the one or more processors, a score corresponding to a machine at the first worksite layout. The method can include selecting, by the one or more processors, a charging location for the machine at the first worksite layout, according to the score and the worksite metric. The method can include transmitting, by the one or more processors, the charging location to the machine and to a charger at the charging location at the first worksite layout.

[0024] FIG. **1** depicts an example system **100** to implement worksite dynamic charging at a worksite. The system **100** can include a data processing system **102**, operation locations **104**A-N (referred to as operation locations **104** herein), and chargers **106** A-N (referred to as chargers **106** herein). The above-mentioned components may be communicably connected to each other through a network **101**. The examples of the network **101** may include, but are not limited to, private or public local area network (LAN), wireless local area network (WLAN), metropolitan area network (MAN), wide area network (WAN), and so forth, which may be used to communicate either via peer-to-peer or via the Internet. The network **101** may include or support both wired and wireless communications according to one or more standards and/or via one or more transport mediums.

[0025] The communication over the network **101** may be performed in accordance with various communication protocols such as Transmission Control Protocol and Internet Protocol (TCP/IP), User Datagram Protocol (UDP), and IEEE communication protocols. In one example, the network **101** may include wireless communications according to Bluetooth specification sets, or another standard or proprietary wireless communication protocol. In another example, the network **101** may also include communications over a cellular network, including, e.g., a GSM (Global System for Mobile Communications), CDMA (Code Division Multiple Access), EDGE (Enhanced Data for Global Evolution) network.

[0026] The system **100** is not confined to the components described herein and may include additional or alternate components, not shown for brevity, which are to be considered within the scope of the embodiments described herein.

[0027] The system **100** can include at least one data processing system **102**. The data processing system **102** can include a physical computer system operatively coupled or that can be coupled with one or more components of the system **100**, either directly or through an intermediate computing device or system. The data processing system **102** can include a virtual computing system, an operating system, and a communication bus to effect communication and processing. The data processing system **102** can include a distributed controller **110**, a central controller **112**, a system processor **114**, and memory **116**.

[0028] The system **100** can include at least one memory **116** within the data processing system **102**. The memory **116** can be coupled to the distributed controller **110**, the central controller **112**, and the system processor **114**. The memory **116** can store data associated with the system **100**. The memory **116** can include one or more hardware memory devices to store binary data, digital data, or the like. The memory **116** can include one or more electrical components, electronic components, programmable electronic components, reprogrammable electronic components, integrated circuits, semiconductor devices, flip flops, arithmetic units, or the like. The memory **116** can include at least one of a non-volatile memory device, a solid-state memory device, a flash memory device, and a NAND memory device. The memory **116** can include one or more addressable memory regions disposed on one or more physical memory arrays. A physical memory array can include a NAND gate array disposed on, for example, at least one of a particular semiconductor device, integrated circuit device, or printed circuit board device.

[0029] The distributed controller **110**, the central controller **112**, and the system processor **114** (referred to as the controllers) of the system **100** can execute one or more instructions associated with the data processing system **102**. The controllers can include an electronic processor, an integrated circuit, or the like including one or more of digital logic, analog logic, digital sensors, analog sensors, communication buses, volatile memory, nonvolatile memory, and the like. The controllers can include, but are not limited to, at least one microcontroller unit (MCU), microprocessor unit (MPU), central processing unit (CPU), graphics processing unit (GPU), physics processing unit (PPU), embedded controller (EC), or the like. The controllers can include memory operable to store or storing one or more instructions for operating components of the controllers and operating components operably coupled to the controllers. For example, the one or more instructions can include one or more of firmware, software, hardware, operating systems, embedded operating systems. The controllers or the data processing system **102** generally can include one or more communication bus controllers to effect communication between the controllers and the other elements of the data processing system **102**.

[0030] The system **100** can include one or more chargers **106**A-N (referred to as chargers **106** herein) at a worksite. The chargers **106** can be connected to one or more power sources, such as electrical outlets, generators, or power supply units, among others. For example, a charger **106**A can be connected to a microgrid to receive electrical power. The chargers **106** can provide electrical energy to batteries, electrical tools, and/or machines **108**A-N (referred to as the machines **108** herein). For example, a power tool can connect to a charger **106**A to recharge a battery of the power tool. The chargers **106** can charge the machines **108** during a workday or outside of operational hours. For example, a machine **108**A can charge at a charger **106** overnight at a worksite.

[0031] The system **100** can include one or more operation locations **104**. The operation location **104** can be or include a designated or specific area within the worksite, where a plurality of activities corresponding to the work environment take place. For example, at a construction site, an operation location **104** can be an excavation site or a building assembly. In another example, at mining site, an operation location **104** can be an area or location where mineral/material/fluids/ore extraction occurs. Each operation location **104** can include one or more machines **108**. For example, referring briefly to FIG. **2**, an operation location **104**B can include machine **104**B. In another example, an operation location **104**A can include machines **108**A and **108**C. In yet another example, an operation location **104**C may not include any machines **108**.

[0032] Referring back to FIG. **1**, the system **100** can include at least one distributed controller **110** within the data processing system **102**. In some embodiments, each machine **108** at the worksite can include the distributed controller **110**. For example, a first machine **108**B can include a first distributed controller **110** and a second machine **108**F can include a second distributed controller **110**. The distributed controller **110** can gather data associated with the worksite. The distributed controller **110** can be configured to communicate with one or more sensors to collect the data associated with the worksite. For example, one or more sensors can gather data regarding equipment usage, fuel consumption, project progress, and environmental conditions. In some arrangements, the distributed controller **110** can communicate with one or more devices to gather the data. For example, a distributed controller **110** can communicate with various devices, such as GPS devices, RFID tags, sensors on machines **108**, and cameras to gather data.

[0033] The distributed controller **110**, the central controller **112** or the system processor **114** can analyze the data and convert the data into a usable format. For example, a camera can transmit a video of the movement of a machine **108**A, during a workday, to a distributed controller **110**. The distributed controller **110** can process the video to extract relevant data (e.g., time of the movement, distance traveled of the machine, etc.). In another example, a sensor on a machine **108**A can report usage of the machine **108**A to a distributed controller **110**. The distributed controller **110** can use the usage of the machine **108**A in one or more formulas to convert the data into a metric. The distributed controller **110** can store the metric and the data in the memory **116** for use during a future time period. For example, the distributed controller **110** can store a usage for a machine **108**A in memory **116**. The system processor **114** can extract the usage for the machine **108**A to form a schedule for the machine **108**A.

[0034] The distributed controller **110**, the central controller **112**, or the system processor **114** can calculate the metric for the worksite layout. Referring now to FIG. **1** and FIG. **2**, FIG. **2** depicts an example worksite layout **200**. The worksite layout **200** can include one or more operation locations **104**, chargers **106**, machines **108**, paths **202**A-N (referred to as "paths **202**" or "path **202**" herein) and charging locations **204**A-N (referred to as "charging locations **204**" or "charging location **204**" herein). The worksite layout **200** is shown as an example and can change daily, weekly, or monthly among others. For example, in worksite layout **200** on a first date, a machine **108**B can be in an operation location **104**B, whereas in worksite layout **200** on a subsequent or different date, a machine **108**A can be in the operation location **104**B.

[0035] The worksite layout **200** can include at least one path **202**. The path **202** can be or include a route from the machines **108** to the charging locations **204**. For example, a path **202**G can lead a machine **108**F to the charging location **204**B. Some paths **202** can be a route between the operation location **104** and the charging location **204**. For example, a path **202**A an connect an operation location **104**B with a charging location **204**A. Some paths **202** can be a route from the machine **108** to the charger **106**. For example, a path **202**H can lead a machine **108**G to a charger **106**K. The worksite layout **200** can include at least one charging location **204**. The charging locations **204** can include one or more chargers **106**. For example, a charging location **204**B can include one or more chargers **106**E-J.

[0036] Referring again to FIG. **1**, as noted above, the distributed controller **110**, the central controller **112**, or the system processor **114** can calculate metric(s) for the worksite layout. In some embodiments, the metric can include productivity metrics, resource utilization, cost, safety metrics, and environment impact, among others. For example, a productivity metric can be based on the amount of work completed, time taken to complete the amount of work, and the number of resources used during the time. In another example, resource utilization can indicate how efficiently equipment and labor are used at the worksite. In yet another example, an environmental impact can access an impact of the worksite on the surrounding environment. The metric can be a percentage, a rate, a numerical value, or on-time task completion. For example, a metric indicate a percentage of budget used during a workday at a worksite.

[0037] The distributed controller **110** or the central controller **112** can base the metric on the cost at the worksite and a historic worksite metric. The distributed controller **110** or the central controller **112** can use one or more formulas or algorithms to combine the cost and the historic worksite metric. For example, a formula can be a cost at the worksite multiplied by a historic worksite metric. In another example, a formula for the cost can be units produced divided by hours worked multiplied by the hourly rate. In yet another example, a formula for a historic worksite metric can be based on a summation of previous history worksite metrics.

[0038] The cost at the worksite can be an estimation based on one or more factors during a previous time period (e.g., yesterday, last week, last month). The one or more factors can include the cost of the worksite during a previous time period and an amount of energy spent for the respective machine **108** to reach the respective charger **106** at the worksite layout **200** during the previous time period. For example, a cost at a worksite today can be based on a difference between the cost at the worksite yesterday and an amount of energy spent by a machine **108**A to reach a charger **106**A at the worksite layout **200** yesterday. In some arrangements, the cost at the worksite can be an estimation based on the project. The project can require the worksite layout **200** to be the same over an extended period of time (e.g., weeks, months, years, etc.). For example, the worksite layout **200** can be the same from Oct. 22, 2022-Mar. 15, 2023. A cost at the worksite can a value based on the stages of the project from Oct. 22, 2022-Mar. 15, 2023. Thus, within the stages of the project, the cost at the worksite can be the same.

[0039] The distributed controller **110** or the central controller **112** can calculate the historic worksite metric, based on a difference between a total amount of work and an amount of productive amount of work at a worksite layout **200** of the previous time period. The total amount of work can be based on one or more factors such as a collection of resources used, an amount of energy spent by the machines **108** at the one or more operation locations **104**, an amount of energy spent by the machines **108** at the worksite, an amount of energy spent by the chargers, progress toward the completion of a project for the worksite, or total cost of the worksite, among others. For example, a total amount of work can be based on an amount of energy spent by a machine **108**A at an operation location **104**A. The amount of productive work at the worksite layout **200** of the previous time period can be based on the same factors as the total amount of work, although the factors must progress toward the completion of the project at the worksite. For example, a machine **108** traveling to the incorrect operation location **108**A can add to a total amount of work but may not contribute to an amount of productive work. Conversely, a machine **108** traveling to the correct operation location **108**A can add to a total amount of work and to an amount of productive work.

[0040] In some arrangements, the distributed controller **110** or the system processor **114** can calculate new charging locations **204** using a linear optimizer. The linear optimizer can be a mathematical algorithm to find an optimal outcome. The linear optimizer can be configured to compute or execute an objective function (e.g., worksite cost, historic worksite metric) to estimate necessary parameters for a future time period. For example, the liner optimizer can be configured or executed to find an optimal worksite cost (e.g., using a cost function). In another example, a linear optimizer can be used to find an optimal historic worksite metric. The new charging location **204** can be for a future time period. The distributed controller **110** or the system processor **114** can use the historic work metric and worksite cost to find new charging location **204** to reduce the worksite cost and improve the historic worksite metric. For example, a new charging location **204**C can be closer to operational location **104**A to reduce the worksite cost for movement of machine **108**D. In another example, a new charging location **204**D can be closer to operational location **104**B and operational location **104**A to improve a historic worksite metric of a worksite **200**. The distributed controller **110** or the system processor **114** can determine a new worksite layout **200** based on the new charging locations **204**.

[0041] The distributed controller **110** or the central controller **112** calculate a score corresponding to the one or more machines **108**. The score can be calculated using one or more factors according to the work layout. The factors can include at least one of distance from the respective machine **108** to the respective charger **106**, amount of energy consumed by the respective machine **108** to reach the respective charger **106**, charge power needed, charger **106** capability, charge time needed, charger **106** uptime, time to charge, and cost of energy at that time. For example, the score can be based on a distance from a machine **108**A to a charger **108**A, a time for the machine **108**A to charge, and the charger **106** capability. In another example, the score can be based on an amount of energy consumed by a machine **108**A to reach the charger **106**A, the charger **106** capability, and a charge time needed for the machine **108**A.

[0042] The distributed controller **110** can calculate the score corresponding to the machine, according to a machine criticality, a queue, a charge connection success rate, and a time to charge. The distributed controller **110** can receive the machine criticality at the worksite layout **200** corresponding to the machine **108**. The machine criticality can indicate an impact relating to downtime for the machines **108**. For example, a machine **108**A can transmit a machine criticality to a distributed controller **110** to identify one or more conditions in which the machine **108**A has experienced downtime (e.g., time in which the machine **108**A is idle, not being used, inactive, disabled, lacks power, etc.).

[0043] The distributed controller **110** can receive the queue at the charger **106** from the central controller **112**. The queue at the charger **106** can indicate one or more machines **108** requesting to charge at the charger. For example, a queue at charger **106**B can include a first machine **108**B, a second machine **108**E, a third machine **108**D and a fourth machine **108**C. The distributed controller **110** can receive the charger connection success rate from the central controller **112**. The charger connection success rate can indicate that one or more chargers **106** are functioning properly. In one example, a charger **106**C can have a charger success rate of 85%, indicating that the charger **106**C can be connected to one or more machines **108**, but may not be successfully connected to at least one machine **108**. In some embodiments, the charger success rate may indicate a likelihood of successful connection to a machine **108** (e.g., based on how many instances in which the charger **106** was unable to successfully connect for charging). The distributed controller **110** can receive a time to charge based on a cost of energy from the central controller **112**. For example, to reduce a worksite cost, a central controller **112** can send a time to charge to a distributed controller **110**.

[0044] Referring now to FIG. **1** and FIG. **3**, FIG. **3** depicts an example data processing system **102** to implement geography optimization at the worksite. The central controller **112** can include a location selector **308**, a path calculator **310**, a load calculator **312**, and an energy calculator **314**. The location selector **308** can select the charging location **204** based on various factors described herein. The factors can include but are not limited to the worksite layout **200**, battery metrics, movement of machines, and movement of materials among others. The path calculator **310** can calculate a path **202** corresponding to the worksite layout **200**. The path calculator **310** can calculate the path **202** based on the worksite layout **200**, location of machines **108**, location of chargers **106**, location of the operation locations **104**, among others. The load calculator **312** can calculate a load for the machines **108**. For example, a load calculator **312** can calculate the loads of machine **108**A, machine **108**B, and machine **108**C. The load can be calculated using a weight for the respective machines, energy spent of a machine, or a terrain for a worksite layout **200**. In some arrangements, the load can impact the score for the machines **108**. For example, a load calculator **312** can calculate a higher load for a machine **108**A. Thus, the score for the machine **108**A can increase because of the higher load.

[0045] The energy calculator **314** can calculate a predicted energy usage or a continuous energy usage. The predicted energy usage can use a load predictor **316** of the system processor **114**. The load predictor **316** can predict a load for the machines **108** at the start of the day at the worksite layout **200**. For example, a load predictor **316** can estimate a load for a bulldozer at a worksite **200** (e.g., based on one or more shifts or related stored information identifying predicted or expected usage of the bulldozer at the worksite **200** within a particular time period). Thus, an energy calculator **314** of the central controller **112** can use the load to calculate a predicted energy usage. The predicted energy usage can use a charging scheduler **318** of the system processor **114**. The charging scheduler **318** can leverage shift operations **324** in memory **116** to determine a schedule to charge the machines **108** based the predicted load for the machines **108**. The shift operations **324** can correspond to a plurality of tasks for the machines **108** at the worksite. For example, a machine **108**A may have shift operations **324** corresponding to excavation and transportation of materials. Based on the shift operations **324**, a charging scheduler **318** can determine a schedule to charge the machine **108**A.

[0046] The location selector **308** of the central controller **112** can select a charging location **204** for the respective machine **108** at the worksite layout **200**. The charging location **204** can be a location in which one or more chargers **106** are located. For example, a charging location **204** can include a charger **106**A. In another example, a charging location **204**A can include a first charger **106**A and a second charger **106**B. The location selector **308** may be configured to select, compute, or otherwise determine the charging location **204**, based on the calculated score and the calculated worksite metric. For example, the location selector **308** can use a score and a worksite metric for a machine **108**E to select charger **106**A or charger **106**B. In another example, a location selector **308** of the central controller **112** can use a score and a worksite metric for a machine **108**B to select charger **106**A. In yet another example, a location selector **308** of the central controller **112** can use a score and a worksite metric for a machine **108**F to select charger **106**C.

[0047] The location selector **308** can select the charging location **204** based on battery metrics of the respective machine **108** and an availability of chargers **106**. The battery metrics can indicate a state of charge, a state of health, a voltage, a charging time, or maintenance parameters. For example, a very low state of charge can indicate that a machine **108**A can only travel to the closest charger **106**A. In another example, a high state of charge can indicate that a machine **108**A can travel to a charger **106**A that is farther away from the machine **108**A. In yet another example, a machine **108**A with maintenance parameters can indicate that the machine **108**A will need maintenance prior to traveling to a charger **106**A. In some arrangements, the battery metrics can impact the score for the respective machine **108**A. For example, a machine **108**A having batteries which are outputting a low voltage (e.g., indicating that the batteries are close to depleted) can have an increased score. In another example, a machine **108**A with a low charging time can have a reduced score.

[0048] The location selector **308** can select the charging location **204** based on the worksite layout **200**. In some embodiments, the location selector **308** may be configured to maintain or otherwise access data corresponding to the worksite layout **200**. The location selector **308** may, for example, access a map (e.g., geographic and/or terrain map) which includes data corresponding to locations of various components or elements of the worksite. As, for example, machines **108** move about the location, the location selector **308** may be configured to update the map (e.g., in real-time or substantially real-time) according to the changes in location. In some instances, a worksite layout **200** can include a terrain that can cause the one or more machines **108** to use a higher amount of energy travel to the one or more chargers **106**. For example, a charger **106**K can be on top of a hill and can be the closest charger **106** to a machine **108**G. Thus, the location selector **308** can select a charger **106**E to reduce an amount of energy spent by the machine **108**G when traveling to the selected charger **106**E. The worksite layout **200** can include one or more charging locations **204** with a high availability of chargers **106**. Thus, the location selector **308** can select the charging location **204** based on the worksite layout **200** and the availability of the chargers **106**. For example, a charging location **204**A can have a high availability of chargers **106**A-K (e.g., based on none of the chargers **106**A-K being currently used to charge a machine **108**), whereas a charging location **204**B can have a low availability of chargers **106**L-O (e.g., based on most [or all] chargers being currently used to charge a machine **108**). Therefore, a location selector **308** of the central controller **112** can select the charging location **204**A, even though fewer chargers are present at charging location **204**A relative to charging location **204**B.

[0049] The location selector **308** can select the charging location **204** based on a movement of materials and a movement of the machines **108**. The materials for the worksite layout **200** can change based on the worksite layout **200**. For example, materials for a demolition worksite layout **200** can be different from the materials of an excavation worksite layout **200**. In some arrangements, the materials can move between operation locations **104** along one or more paths **202**. For example, if materials need to be moved from operation location **104**A to operation location **104**B along path **202**G, path **202**B, and path **202**A, a location selector **308** of the central controller **112** can select a charging location **204**B or charging location **204**A along the respective paths **202**. The movement of the machines **108** can change based on the worksite layout **200** and the operation locations **104**. For example, a machine **108**G may travel to operation location **104**C along path **202**H, path **202**E, path **202**G, path **202**B, and path **202**C respectively. Thus, a location selector **308** of the central controller **112** can select a charging location **204** along the path **202**.

[0050] The central controller **112** can transmit the charging location **204** to the distributed controller **112**. The distributed controller **110** can include at least one charger selector **302**, path selector **304**, and machine processor **306**. The charger selector **302** can determine the charger **106** according to the charging location **204**. For example, a location selector **308** can select charging location **204**A at a worksite **200**. The charger selector **302** can leverage the availability of chargers **106**A-D at the charging location **204**A to select charger **106**B for a machine **108**A. In another example, a location selector **308** can select charging location **204**A at a worksite **200**. The charger **302** can leverage the availability of chargers **106**A-D at the charging location **204**A to select a first charger **106**A or a second charger **106**B for a machine **108**A. In some arrangements, the charger selector **302** can determine the charger **106** not according to the charging location **204**. For example, a charging selector **302** of a distributed controller **110** can leverage the availability of chargers **106**A-K at the worksite **200** to select charger **106**K for a machine **108**A.

[0051] The path selector **304** can determine the path **202** to the charger **106** according to the worksite layout **200**. To determine the path **202** to the charger **106**, the path selector **204** can use the path calculator **310** of the central controller **112**. The path selector **304** can select the path **202** which is the best path **202** for the machines **108** to reach the charger **106**. For example, a path calculator **310** can calculate paths **202**A-H for a worksite layout **200**. A path selector **304** of a machine **108**A can select path **202**E, path **202**G, and path **202**F to reach charger **106**K.

[0052] The machine processor **306** can transmit a signal to the machines **108**. In some embodiments, such as where the distributed controller **110** is provided at each machine **108**, the signal can be sent directly to the machine **108**. Otherwise, the signal is sent from the distributed controller **110** over the network **101** to the machine **108**. The signal can indicate or trigger the machines **108** to follow the path **202** to the charger **108**. For example, a machine processor **306** can transmit a signal to a machine **108**D. In response to receiving the signal, the machine **108**D may follow path **202**D to a charger **106**E. The signal can include information corresponding to load cycles of the machine **108** and energy generation/storage at the charger **106**. For example, a signal from the machine processor **306** can indicate load cycles for a machine **102**G to travel to charger **106**K along path **202**H. Furthermore, the signal can indicate an amount of energy stored at the charger **106**K.

[0053] The machine processor **306** can transmit the score to the selected charger **106**. The selected charger **106** can receive a plurality of scores from the machine processors at the machines **108**. For example, a first machine **108**A, a second machine **108**B, and third machine **108**C can each send a score to charger **106**J. The distributed controller **110** can receive an acceptance of the score by the chargers **106** and transmit a signal to the machines **108**. The acceptance of the score can indicate that the machines **108** can proceed to the chargers **106**. For example, a distributed controller **100** can receive an acceptance of a score from charger **106**K and the distributed controller **110** can send a signal to allow a machine **108**G to proceed to the charger **106**K. In some arrangements, the chargers **106** can reject the score of the machines **108**. For example, if a queue of a charger **106**A is too larger, the charger **106**A can reject a score of a machine **108**A.

[0054] The central controller **112** can broadcast charger information of the acceptance of the score to the plurality of machines **108**. The charger information can indicate an availability status and power availability of microgrid assets **322** in memory **116**. The microgrid assets **322** can correspond to the amount of power available at a power grid from the worksite **200**. The microgrid assets **322** can include utilities, generator sets, and renewable energy sources. The availability status can indicate how available the chargers **106** are throughout the workday. For example, a first charger **106**A with a low availability indicates that the first charger **106**A is not available during a workday, whereas a second charger **106**B with a high available indicate that the second charger **106**B is available during the day. In response to broadcasting the charger information, the plurality of machines **108** can transmit a plurality of bids to the central controller **112**. Each bid in the plurality of bids can include a request to charge the corresponding machine **108**. For example, a plurality of machines **108**A-N can send a plurality of bids to a central controller **112**. The central controller **112** can determine one bid in the plurality of bids corresponding to the charger information, to charge the corresponding machine **108** at the respective charger **106**. For example, a plurality of machines **108**A-N can send a plurality of bids to a central controller **112** for a charger **106**C. The central controller **112** can determine that a first machine **108**C of the plurality of machines **108** can proceed to the charger **106**C.

[0055] The distributed controller **110** can transmit the charging location to the respective machine **108**. Each machine **108** in the one or more machines **108** can include a distributed controller **110** to allow for communication over the network **101**. The distributed controller **110** can transmit the charging location using the network **101** or using a hard wired connection with the machine **108**. For example, a distributed controller **110** can transmit a charging location to a machine **108** through a hardwired connection with the machine **108**A. In another example, a distributed controller **110** of a first machine **108**A can transmit a charging location to a distributed controller **110** of a second machine **108**B to broadcast the charging location to one or more machines **108** at the worksite.

[0056] The distributed controller **110** can transmit the charging location to the respective charger **106**. The distributed controller **110** can transmit the charging location to the one or more chargers **106** using the network **101**. For example, a distributed controller **110** of a machine **108**E can transmit a charging location to chargers **106**A-D. In another example, a distributed controller **110** of a machine **108**G can transmit a charging location to a charger **106**K. The worksite layout **200** can change daily; therefore, the operation locations **104**A-N, the chargers **106**A-N, and the charging locations can vary per day. For example, a first worksite layout **200** on Jan. 8, 2023, can be different from a second worksite layout **200** on Jan. **6**, **2023**. In another example, a first worksite layout **200** on Jan. 8, 2023, can be the same as a second worksite layout **200** on Jan. 7, 2023.

[0057] The central controller **112** can receive information between the one or more machines **108** and the one or more chargers **106**. The information can indicate the battery metrics of the one or more machines **108** and the charger availability of the one or more chargers **106**. For example, a machine **108**A can send battery metrics to a central controller **112** and a charger **106**K can send a charger availability to the central controller **112**. Using the information, the path calculator **310** can calculate a best path **202** for the machine **108**A to reach the charger **106**K.

[0058] FIG. **4** depicts a **400** method to implement worksite dynamic charging for the worksite and microgrid. The method **400** can be performed by, using, or for system **100**. The method **400** can include calculating a worksite metric for a worksite layout (e.g., worksite layout **200**) at step **405**. The method **400** can include calculating a score corresponding to a machine (e.g., machines **108**A-N) at the worksite layout at step **410**. The method **400** can include selecting a charging location (e.g., charging locations **204**A-N) for the machine at the worksite layout at step **415**. The method **400** can include transmitting the charging location to the machine and to a charger (e.g., chargers **106**A-N) at the charging location at step **420**.

[0059] At step **405**, the method **400** can include calculating, by the central controller **112**, the worksite metric for the worksite layout. The worksite metric can be based on a worksite cost and a historic worksite metric during a previous time period. The worksite layout can change per day, week, month, or year among others. Prior to calculating the worksite metric, the method **400** can include calculating, by the central controller **112**, the worksite cost, based on a difference between a total cost and an amount of energy spent by the machine to reach the charger at the second worksite layout, during the previous time period and calculating, by the central controller **112**, the historic worksite metric based on a difference between a total amount of work and an amount of productive work at the second worksite layout, during the previous time period. For example, the central controller **112** can access memory **116** to receive worksite costs and historic worksite metrics during a previous time period. The central controller **112** can calculate the worksite metric based on the worksite cost and historic worksite metric to make a prediction for the worksite layout **200**.

[0060] At step **410**, the method **400** can include calculating, by the system processor **114** or the central controller **112**, the score corresponding to the machine at the worksite layout. Prior to calculating the score, the method **400** can include receiving, by the distributed controller **110** or the machine processor **306**, a machine criticality at the first worksite layout corresponding to the machine, a queue at the charger, a charger connection success rate, and a time to charge based on a cost of energy from a central controller. The system processor **114** or the central controller **112** may calculate the score corresponding to the machine, according to the machine criticality, the queue, the charge connection success rate, and the time to charge. The distributed controller **110** can send the score to the charger and receive an acceptance of the score by the charger. The method **400** can include transmitting, by the distribute controller **110**, a signal to the machine. The signal can direct the machine to proceed to the charger of the received acceptance. For example, a distributed controller **100** can receive an acceptance of a score from charger **106**K and the distributed controller **110** can send a signal to allow a machine **108**G to proceed to the charger **106**K.

[0061] The method **400** can include broadcasting, by the central controller **112**, charger information of the charger of the received acceptance to a plurality of machines. The charger information can indicate availability status and power availability. The method **400** can include receiving, by the central controller, a plurality of bids from the plurality of machines. Each bid in the plurality of bids can indicate battery metrics and a request of the corresponding machine. The method **400** can include determining, by the central controller **112**, one bid in the plurality of bids corresponding to the charger information to charge the corresponding machine.

[0062] At step **415**, the method **400** can include selecting, by the charger selector **302** of the distributed controller **110** or the central controller **112** a charging location for the machine at the first worksite layout, according to the score and the worksite metric. The charging location can be based on battery metrics of the machine and an availability of chargers. The battery metrics can indicate at least one of a state of charge, a state of health, a voltage, charging time, or safety parameters. Prior to selecting the charging location, the method **400** can include determining, by the location selector **308** of the central controller **112**, the charging location based on the first worksite layout, movement of materials, and movement of the machine. The method **400** can include determining, by the path calculator **310** of the central controller **112**, the charger according to the charging location, and a path (e.g., paths **202**A-N) to the charger according to the worksite layout. The method **400** can include transmitting, by the machine processor **306** of the distributed controller **110**, a signal to the machine. The signal can indicate that the machine follow the path to the charger. The signal can include information corresponding to load cycles and energy generation of the charge based on the previous time period.

[0063] At step **420**, the method **400** can include transmitting, by the distributed controller **110**, the charging location to the machine and to a charger at the charging location at the first worksite layout. The method **400** can include receiving, by the machine processor **306** of the distributed controller **110**, information between the one or more machines and the one or more chargers from a central controller. The information can indicate battery metrics of the one or more machines and charger availability of the one or more chargers. The method **400** can include computing, by the system processor **114** or the central controller **112**, using a linear optimizer, a new charger location for a future time period, according to the worksite cost and the historic worksite metric. In response to computing the new charging location, the method **400** can include determining, by the by the distributed controller **110** or the central controller **112**, a third worksite layout based the new charger location.

INDUSTRIAL APPLICABILITY

[0064] The disclosed embodiments may be application to any geographic optimization based system or solution. For example, the disclosed embodiments may be applicable or applied to a worksite, such as an excavation site, a mining site, a demolition site, or any other type of industrial worksite, charging locations, charging sites, electrical power grid, a machine, such as a bulldozer, a forklift, or any other type of machinery, a generator, a transformer, or any other type of electrical utility. The disclosed embodiments may be applicable to worksites which include various types of heavy machinery designed to execute tasks throughout the worksite to maximize efficiency and productivity while reducing costs at the worksite. The disclosed distributed controller **110** may be provided to increase efficiency by selecting optimal paths **202** for one or more machines **108** to travel to the selected charger **108**A or selected charging location **204**A. For example, because the distributed controller **110** receives calculated metrics (e.g., load metrics, worksite metrics, worksite cost) from the central controller **112**, the distributed controller **110** can organize each machine **108**A in the one or more machines **108** to travel to a respective charger **106**A to reduce the idle times of the machines **108**.

[0065] In various embodiments of the present solution, the distributed controller **110** can communicate with the central controller **112** to use one or more calculated metrics to determine a worksite layout **200** for a future time period. By exchange metrics based on the chargers **106** and the machines **108**, the distributed controller **110** can continuously select the best paths **202**, charging locations **204**, and chargers **106** in each worksite layout **200** during a project or task at the worksite. The worksite layout **200** can change per day, week, month, or year based on the project or the task. This, the distributed controller **110** and the central controller **112** can determine the most optimal worksite layout **200** to reduce cost, maximize productivity, and maximize efficiency at the worksite.

What is claimed:

1. A method, comprising:

calculating, by one or more controllers, a worksite metric for a first worksite layout, wherein the worksite metric is based on a worksite cost and a historic worksite metric during a previous time period, wherein the first worksite layout differs from a second worksite layout;

calculating, by the one or more controllers, a score corresponding to a machine at the first worksite layout;

selecting, by the one or more controllers, a charging location for the machine at the first worksite layout, according to the score and the worksite metric; and

transmitting, by the one or more controllers, a first signal corresponding to the charging location to the machine, and a second signal to a charger at the charging location.

2. The method of claim **1**, wherein calculating the worksite metric for the first worksite layout comprises:

calculating, by the one or more controllers, the worksite cost, based on a difference between a total cost and an amount of energy spent by the machine to reach the charger at the second worksite layout, during the previous time period; and

calculating, by the one or more controllers, the historic worksite metric based on a difference between a total amount of work and an amount of productive work at the second worksite layout, during the previous time period.

3. The method of claim **1**, wherein selecting the charging location is based on battery metrics of the machine and an availability of chargers, wherein the battery metrics indicates at least one of a state of charge, a state of health, a voltage, charging time, or safety parameters.

4. The method of claim **1**, wherein selecting a charging location for the machine at the first worksite layout further comprising determining, by the one or more controllers, the charging location based on the first worksite layout, movement of materials, and movement of the machine.

5. The method of claim **4**, further comprising:

selecting, by the one or more controllers, the charger according to the charging location, and a path to the charger according to the first worksite layout; and

transmitting, by the one or more controllers, the first signal to the machine, wherein the first signal indicates the machine follow the path to the charger, wherein the first signal comprises information corresponding to load cycles and energy generation of the charge based on the previous time period.

6. The method of claim **1**, wherein transmitting, the charging location to the machine and to the charger at the charging location at the first worksite layout further comprising receiving, by the one or more controllers, information between one or more machines and one or more chargers from a central controller, wherein the information indicates battery metrics and charger availability.

7. The method of claim **1**, further comprising:

computing, by the one or more controllers, using a linear optimizer, a new charger location for a future time period, according to the worksite cost and the historic worksite metric; and

determining, by the one or more controllers, a third worksite layout based the new charger location.

8. The method of claim **1**, wherein calculating, the score further comprises:

receiving, by the one or more processors, a machine criticality at the first worksite layout corresponding to the machine, a queue at the charger, a charger connection success rate, and a time to charge based on a cost of energy from a central controller; and

calculating, by the one or more controllers, the score corresponding to the machine, according to the machine criticality, the queue at the charger, the charge connection success rate, and the time to charge.

9. The method of claim **1**, wherein selecting the charging location for the machine at the first worksite layout further comprises:

sending, by the one or more controllers, the score to the charger,

receiving, by the one or more controllers, an acceptance of the score by the charger; and

transmitting, by the one or more controllers, a signal to the machine, wherein the signal directs the machine to proceed to the charger of the received acceptance.

10. The method of claim **9**, further comprising:

broadcasting, by the one or more controllers, charger information of the charger of the received acceptance to a plurality of machines, wherein the charger information indicates an availability status and a power availability;

receiving, by the one or more controllers, a plurality of bids from the plurality of machines, wherein each bid in the plurality of bids indicates battery metrics of a corresponding machine and a request by the corresponding machine; and

determining, by the one or more controllers, one bid in the plurality of bids corresponding to the charger information to charge the corresponding machine.

11. A system comprising:

a plurality of machines comprising one or more first machines at a first operation location and one or more second machines at a second operation location;

a plurality of chargers comprising one or more first chargers at a first charging location and one or more second chargers at a second charging location; and

a data processing system comprising:

memory; and

one or more controllers comprising one or more processors, configured to:

calculate a worksite metric for a first worksite layout, the worksite metric based on a worksite cost and a historic worksite metric during a previous time period, wherein the first worksite layout differs from a second worksite layout;

calculate a score corresponding to a first machine of the plurality of machines at the first worksite layout;

select a charging location for the first machine at the first worksite layout, according to the score and the worksite metric; and

transmit a first signal corresponding to the charging location to the first machine, and a second signal to a charger of the plurality of chargers at the charging location.

12. The system of claim **11**, wherein, when calculating the worksite metric for the first worksite layout, a central controller of the one or more controllers is configured to:

calculate the worksite cost, based on a difference between a total cost and an amount of energy spent by the first machine to reach the charger at the second worksite layout, during the previous time period; and calculate the historic worksite metric based on a difference between a total amount of work and an amount of productive work at the second worksite layout, during the previous time period.

13. The system of claim **11**, wherein selecting the charging location is based on battery metrics of the machine and an availability of chargers, the battery metrics indicates at least one of a state of charge, a state of health, a voltage, charging time, or safety parameters.

14. The system of claim **11**, wherein, when selecting a charging location for the first machine at the first worksite layout, a distributed controller of the one or more controllers is configured to:

determine the charging location based on the first worksite layout, movement of materials, and movement of the machine;

determine the charger according to the charging location and a path to the charger according to the first worksite layout; and

transmit a signal to the first machine, wherein the signal instructs the first machine follow the path to the charger, wherein the signal indicates load cycles and energy generation of the charger based on the previous time period.

15. The system of claim **11**, wherein, when transmitting the first signal corresponding to the charging location to the first machine and the second signal to the charger at the charging location a distributed controller of the one or more controllers is configured to receive information between one or more of the plurality of machines and one or more of the plurality of chargers from a central controller, wherein the information indicates battery metrics and charger availability.

16. The system of claim **11**, wherein the one or more controllers are further configured to:

compute, using a linear optimizer, a new charger location for a future time period, according to the worksite cost and the historic worksite metric; and

determine a third worksite layout based the new charger location.

17. The system of claim **11**, wherein, when calculating the score, a distributed controller of the one or more controllers is configured to:

receive a machine criticality at the first worksite layout corresponding to the first machine, a queue at the charger, a charger connection success rate, and a time to charge; and

calculate the score corresponding to the first machine, according to the machine criticality, the queue at the charger, the charge connection success rate, and the time to charge.

18. The system of claim **11**, wherein, when selecting the charging location for the first machine at the first worksite layout, a distributed controller of the one or more controllers is configured to:

send the score to the charger;

receive an acceptance of the score by the charger;

broadcast charger information of the charger of the received acceptance to the plurality of machines, wherein charger information indicates availability status and power availability;

receive a plurality of bids from the plurality of machines, wherein each bid in the plurality of bids indicates battery metrics of a corresponding machine and a request by the corresponding machine; and

determine one bid in the plurality of bids corresponding to the charger information to charge the corresponding machine; and

transmit the first signal to the first machine, wherein the signal directs the machine to proceed to the charger of the received acceptance.

19. The system of claim **11**, wherein the one or more controllers comprise:

a central controller configured to:

calculate the worksite metric for the first worksite layout; and

calculate the score corresponding to the first machine of the plurality of machines at the first worksite layout; and

a distributed controller configured to:

select the charging location for the first machine at the first worksite layout, according to the score and the worksite metric; and

transmit the first signal corresponding to the charging location to the first machine, and transmit a second signal to the charger of the plurality of chargers at the charging location.

20. A data processing system comprising:

memory; and

one or more controllers comprising one or more processors, configured to:

calculate a worksite metric for a first worksite layout, the worksite metric based on a worksite cost and a historic worksite metric during a previous time period, wherein the first worksite layout differs from a second worksite layout;

calculate a score corresponding to a first machine of a plurality of machines at the first worksite layout;

select a charging location for the first machine at the first worksite layout, according to the score and the worksite metric; and

transmit a first signal corresponding to the charging location to the first machine, and a second signal to a charger of a plurality of chargers at the charging location.

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